EXPERIMENT NO. 5

Maximum Element of an Array with Least Time Complexity

Programming Language: Java

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Aim

To write a Java program that finds the maximum element of an array in the least possible time complexity.

Question

Write a program for the below test cases with the least time complexity.

Test Case 1

Input: {1, 2, 3, 4, 5}

Expected Output: 5

Test Case 2

Input: {7, 7, 7, 7, 7}

Expected Output: 7

Test Case 3

Input: {-10, 2, 3, -4, 5}

Expected Output: 5

Algorithm

Step 1: Start.

Step 2: Read the array arr of size n.

Step 3: Set max = arr[0].

Step 4: For i from 1 to n-1, if arr[i] > max then set max = arr[i].

Step 5: Print max.

Step 6: Stop.

Program (Java)

```
public class Main {

    static int findMax(int[] arr) {
        int max = arr[0];
        for (int i = 1; i < arr.length; i++) {
            if (arr[i] > max)
                max = arr[i];
```

```
        }
        return max;
    }

    public static void main(String[] args) {
        int[] t1 = {1, 2, 3, 4, 5};
        int[] t2 = {7, 7, 7, 7, 7};
        int[] t3 = {-10, 2, 3, -4, 5};

        System.out.println("Test Case 1 Output: " + findMax(t1));
        System.out.println("Test Case 2 Output: " + findMax(t2));
        System.out.println("Test Case 3 Output: " + findMax(t3));
    }
}
```

Output

```
Test Case 1 Output: 5
Test Case 2 Output: 7
Test Case 3 Output: 5

=== Code Execution Successful ===
```

Complexity Analysis

Time Complexity: O(n) - a single pass; this is optimal because every element must be examined at least once.

Space Complexity: O(1)

Result

Thus the program was executed successfully and the output was verified.